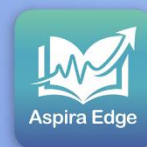




2026-27

CONCEPTS | CLINICALS | MCQS



ANATOMY

- Master the Structure. Conquer the Questions.
- High-yield diagrams, nerve lesions & clinical correlations.
- PYQ-focused integration for rapid recall.



DR. KEYUR GANDHI

DR. JITENDRA

ANATOMY

ORDER OF FORMATION OF GERM LAYERS- Endoderm > Mesoderm>Ectoderm

Days of fertilization	Process
Day 1	One celled stage (zygote)
Day 2	Two celled stage
Day 3	Morula
Day 4	Advanced morula >32 days cells → enters the uterine cavity
Day 6	blastocyst ; Implantation
Day 12	Uteroplacental circulation starts Fetal placental circulation starts
3rd week	Fetal placental circulation starts

After day five, implantation begins and is completed by day twelve.

In the 3rd week of development, there is formation of 3 germ layers. This is called **GASTRULATION**.

These 3 germ layers are formed from **EPIBLAST**

Q.Which is the first germ layer to form - **ENDODERM**

Q.Mesoderm is not present in which 2 layers - **CLOACAL MEMBRANE**
BUCCOPHARYNGEAL MEMBRANE

Q. Name the 4th germ layer- **NEURAL CREST CELL**

TYPES OF CARTILAGE

1.HYALINE CARTILAGE

MC type of cartilage

Location- present between tips of ribs and bones of sternum, covering bone surface at synovial joints.

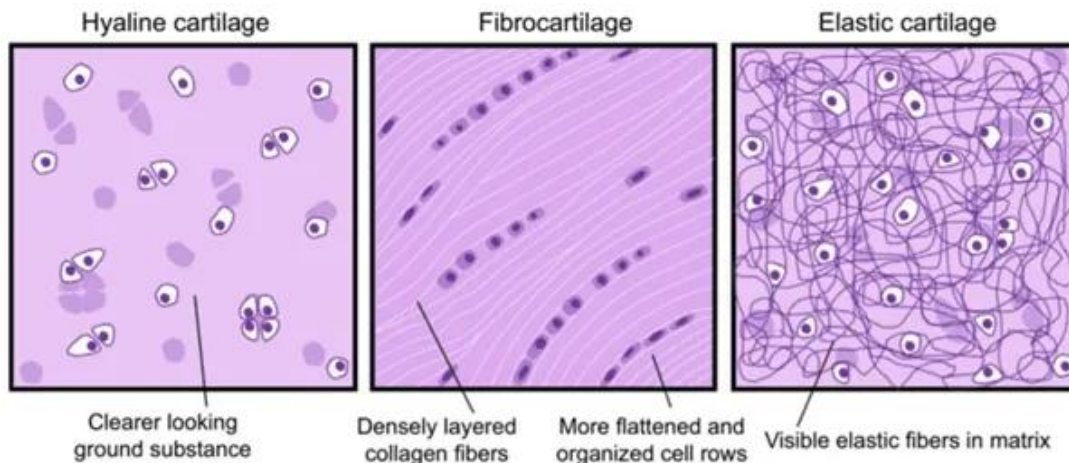
Functions - reduces friction between 2 bony surfaces.

2.ELASTIC CARTILAGE

Location- auricle of external ear, epiglottis, auditory tube, cuneiform cartilages of skin.

3.FIBROCARILAGE

Location- pads within knee joints, between pubic bones of pelvis, intervertebral discs.

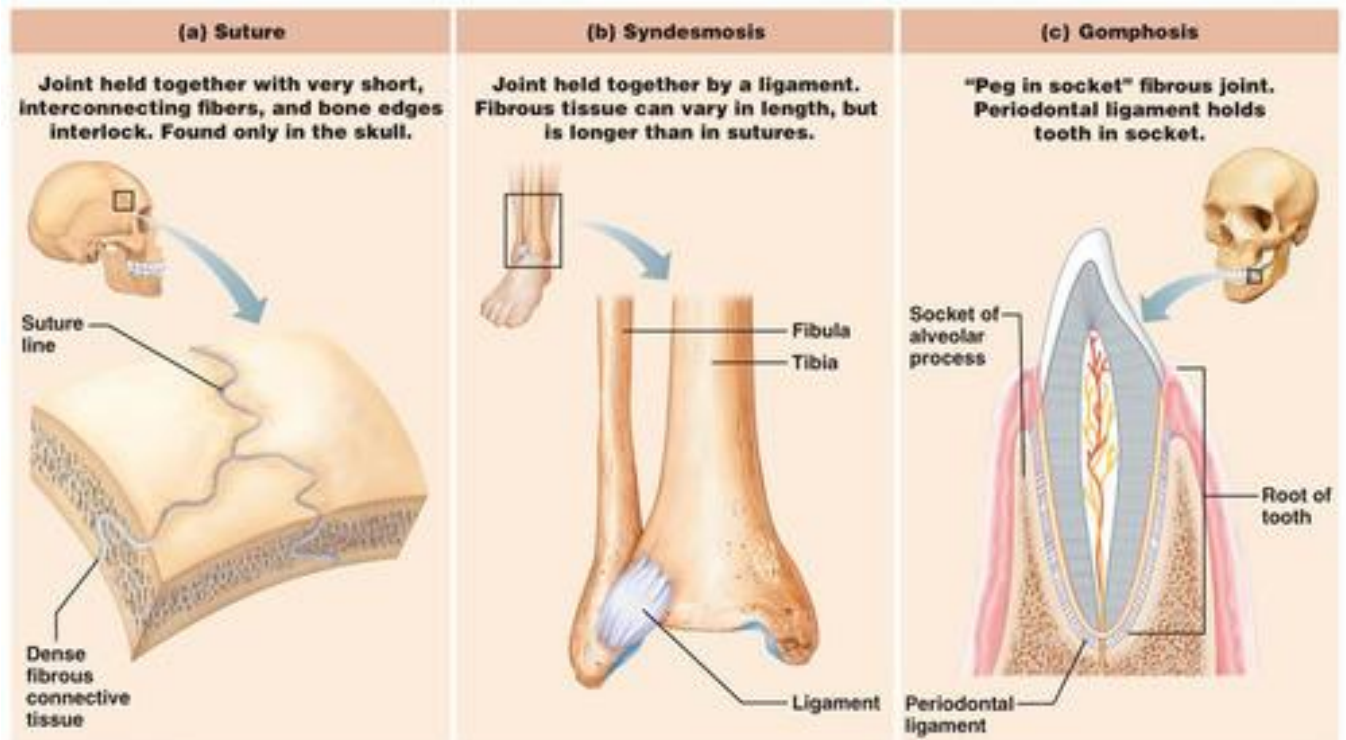


TYPES OF JOINTS

1.FIBROUS JOINTS

aka immovable joints.

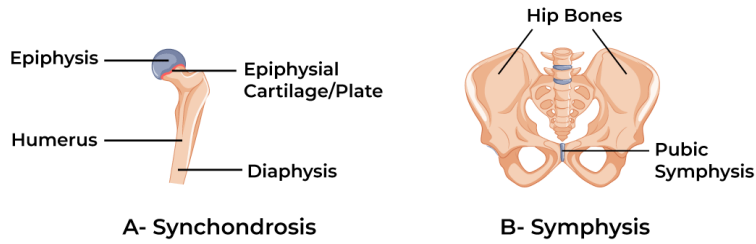
Eg- skull, upper jaw, rib cage, backbone, pelvic bone



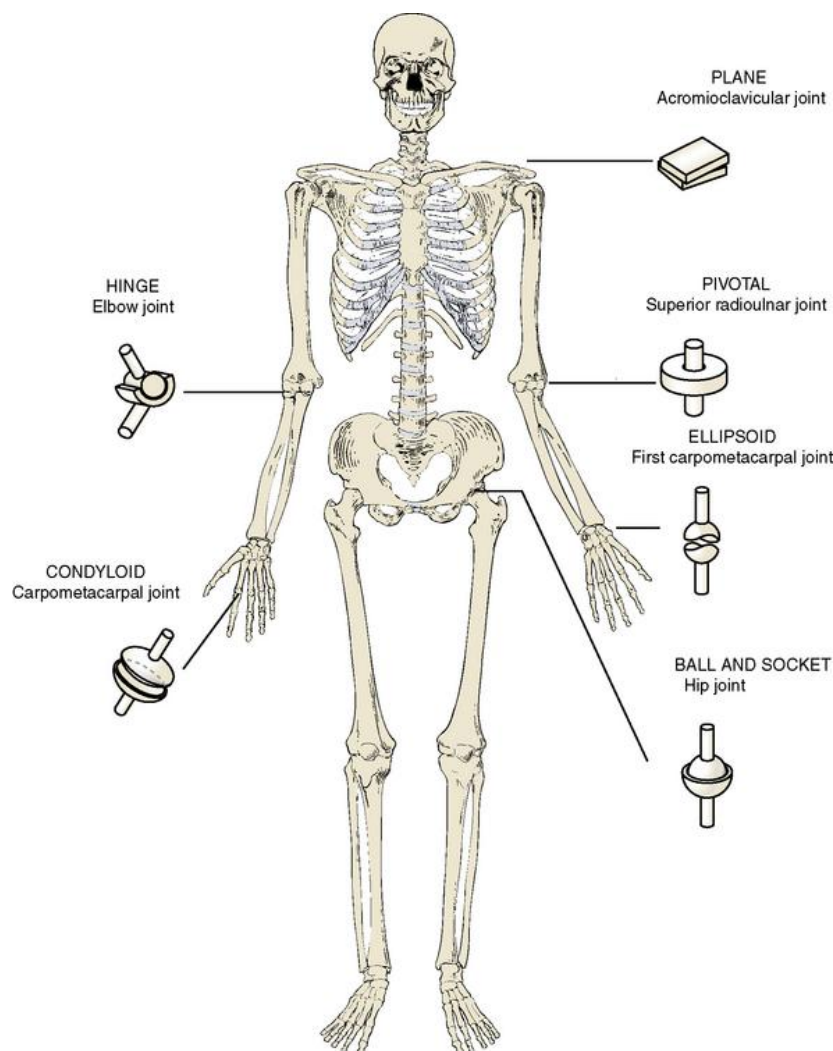
TYPES OF FIBROUS JOINTS

CARTILAGINOUS JOINTS- Partly movable joints

Cartilaginous Joint



SYNOVIAL JOINTS- The synovial joints are the most common type of joint because this joint helps to perform a wide range of motion.



NEUROANATOMY

Q.First fontanelle to disappear- **POSTERIOR FONTANELLE**

Q. Last fontanelle to disappear- **ANTERIOR FONTANELLE**
(closes b/w 7-19 months of age)

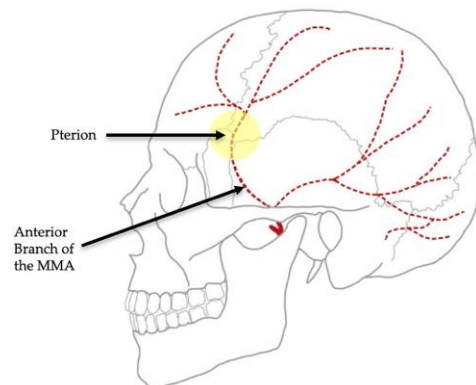
WEAKEST POINT of skull- PTERION

Q: Deep relations of Pterion

- 1) Middle meningeal artery: The pterion overlies the anterior branch of this artery, which enters the skull through the foramen spinosum. Rupture of this artery can lead to an epidural hematoma.
- 2) Middle cerebral artery: This artery runs within the lateral sulcus (Sylvian fissure) directly beneath the pterion.
- 3) Lateral sulcus (Sylvian fissure): The beginning of this major fissure is located deep to the pterion.
- 4) Lesser wing of sphenoid
- 5) Insula: The anterior pole of the insular lobe of the brain is situated deep to the pterion.

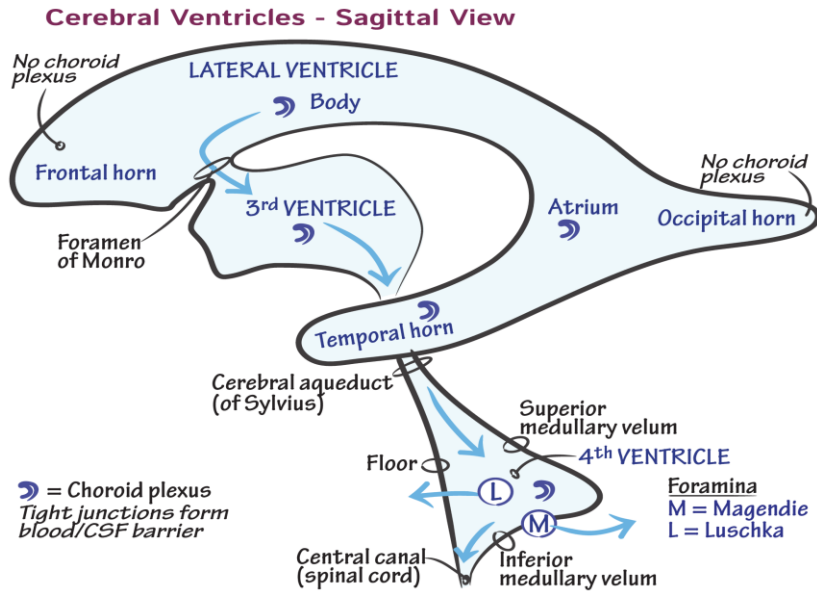
Deep to the pterion there is an anterior branch of MIDDLE MENINGEAL ARTERY.

CSF is produced by- CHOROID PLEXUS



FLOW OF CSF

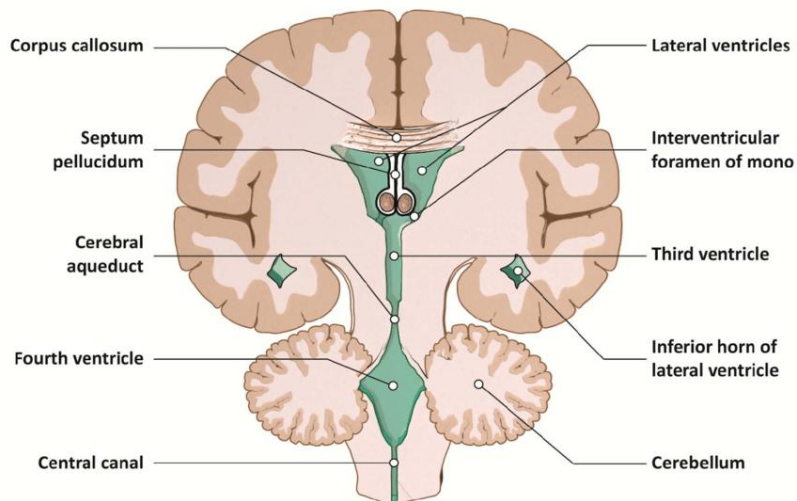
Lateral ventricles → third ventricles → cerebral aqueduct of sylvius → fourth ventricle → central canal



AQUEDUCT STENOSIS

- It is blockage at Sylvius Aqueduct
- Blockage of flow of CSF leads to development of hydrocephalus.
- Lead to internal hydrocephalus
- Makes 3rd and lateral ventricles dilate and forebrain flattening

CORONAL SECTION OF LATERAL VENTRICLES

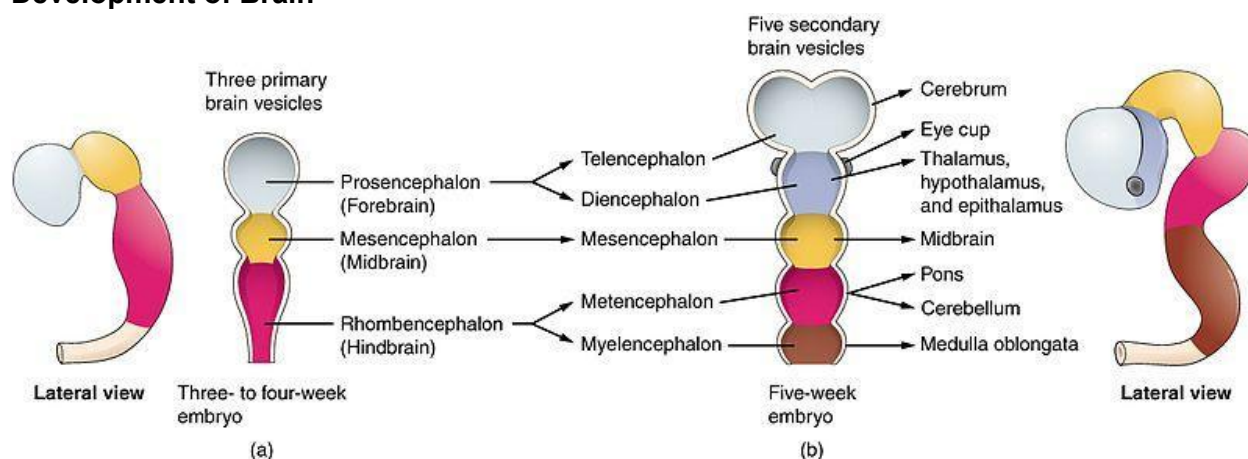


Floor - body of caudate nucleus

Roof - corpus callosum

Medial wall - septum pellucidum

Development of Brain



FUNCTIONAL AREAS

WERNICKE'S APHASIA/ Receptive aphasia / Fluent aphasia

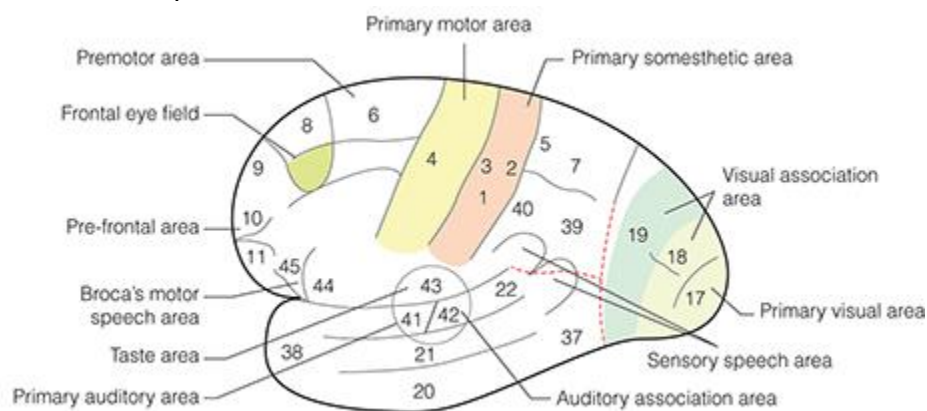
patient is unable to understand written and spoken language

BROCA'S APHASIA/Expressive aphasia/Non fluent aphasia

speech become difficult but patient understands language

ARCUATE FASCICULUS

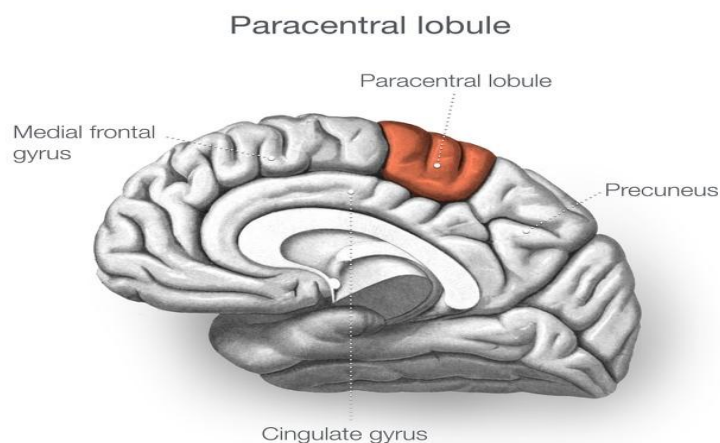
The arcuate fasciculus is a white matter fiber bundle in the brain that connects language areas like Broca's and Wernicke's areas, playing a crucial role in language processing, especially in the left hemisphere



PARACENTRAL LOBULE

The paracentral lobule is a U-shaped region on the medial surface of the brain's cerebral hemisphere that connects the precentral and postcentral gyri.

It is crucial for the motor and sensory functions of the lower limbs and perineum and also plays a role in controlling urination and defecation.



CRANIAL FORAMINA

FORAMEN OVALE - mnemonic- **MALE**

1. Mandibular nerve CN V3
2. Accessory meningeal nerve
3. Lesser petrosal nerve
4. Emissary vein

FORAMEN SPINOSUM - mnemonic - **MEN**

1. Middle meningeal artery and vein (posterior trunk)
2. Emissary vein
3. Nervus spinosus (meningeal branch of mandibular nerve)

FORAMEN LACERUM - mnemonic-

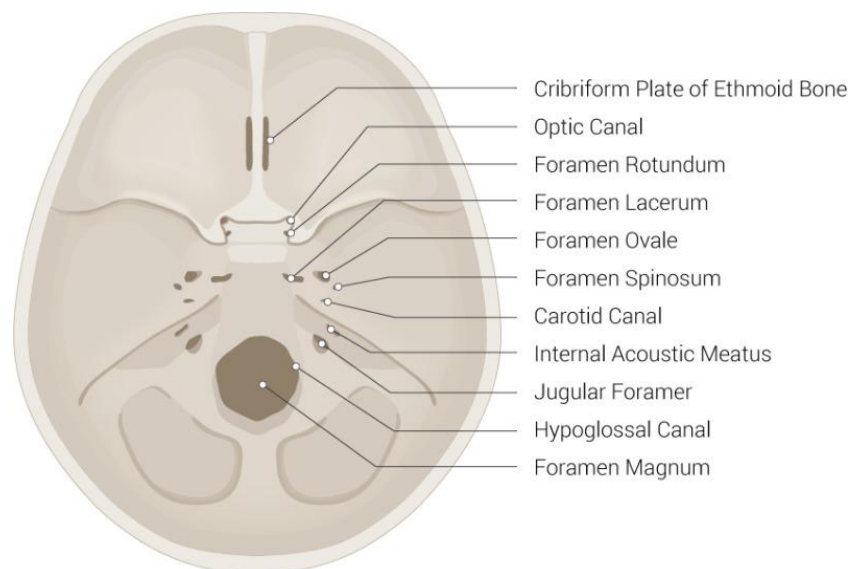
- Structures passing whole length
1. Meningeal branch of ascending pharyngeal artery
 2. Emissary vein

FORAMEN ROTUNDUM

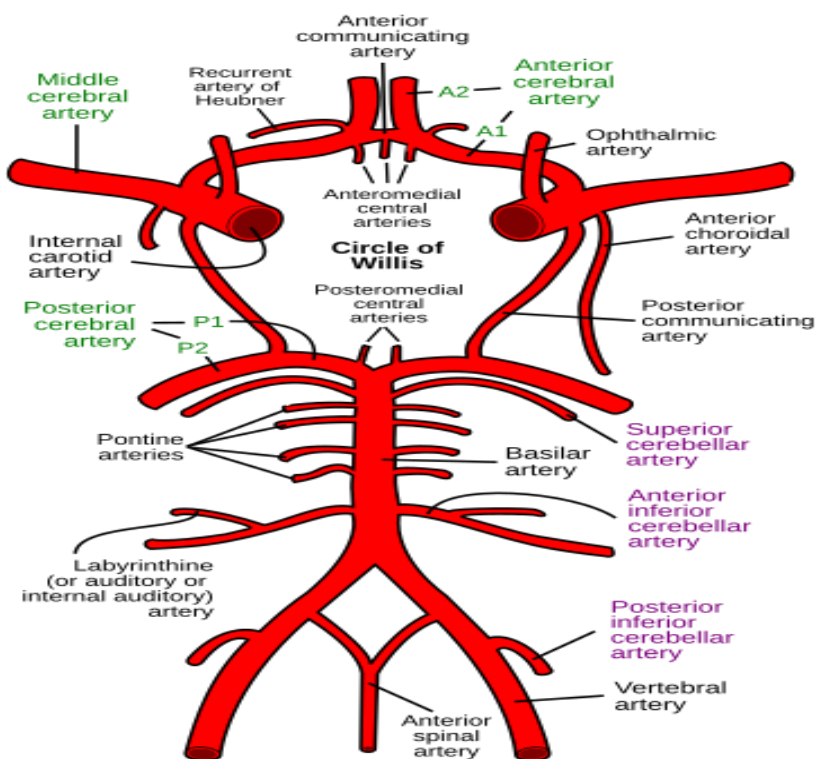
MAXILLARY NERVE (CN V2)

INTERNAL ACOUSTIC MEATUS

1. Facial nerve (CN VII)
2. Vestibulocochlear nerve (CN VIII)



BLOOD SUPPLY OF BRAIN



CIRCLE OF WILLIS

Q. Which artery does not form circle of willis - MIDDLE CEREBRAL ARTERY

Q. MC site for circle of willis - ANTERIOR COMMUNICATING ARTERY
Compression of optic chiasma causes - BITEMPORAL HEMIANOPIA

Q. Aneurysm of posterior cerebral artery presses - 3rd CN

LATERAL MEDULLARY SYNDROME/ WALLENBERG SYNDROME

Artery : PICA

CN involved - HYPOGLOSSAL

Symptoms- N/V, dizziness, giddiness

Difficulty in walking, speaking and eating

Ipsilateral loss of taste sensation

Contralateral loss of pain and temperature

MEDIAL MEDULLARY SYNDROME

Artery : ANTERIOR SPINAL ARTERY

Features :

C/L loss of touch and vibration

12th CN → I/L tongue paralysis

WEBER SYNDROME

Artery: Posterior cerebral artery

Features:

Ipsilateral CN III palsy → ptosis, dilated pupil, eye down & out

Contralateral hemiplegia

MILLARD GUBBLER SYNDROME

Artery : Basilar artery

CN involve → VI and VII involves

Features :

I/L convergent squint

I/L facial paralysis

MENINGES

Duramater, Arachnoid, Piamater

SPACES

Subarachnoid space- between arachnoid and pia mater - CSF (cerebrospinal fluid)

Subdural space- between dura mater and arachnoid

Epidural space- between vertebrae and dura mater

SPINAL CORD

At L3-L4 junction - Lumbar puncture done

1. Iliac crest level

2. Location of umbilicus

Layers pierced by lumbar puncture needle

Skin → fascia → supraspinous ligament → interspinous ligament → ligamentum flavum → dura mater → arachnoid

Level of the spinal cord

Age	Level of the spinal cord
At birth	Ends at the level of 3rd lumbar vertebra
In adult	Terminates at the level of lower border of L1

TRACTS OF SPINAL CORD

1.CORTICOSPINAL TRACT

Origin : motor area 4,6

Somatosensory 3,1,2

Decussation : at lower medulla

2.SPINOTHALAMIC TRACT

Anterior spino thalamic tract - carries → crude, touch and pressure

Lateral spino thalamic tract - carries → pain and temperature

First order neuron - DRG (dorsal root ganglion)

Second order neuron - substantia gelatinosa

Third order neuron - thalamus

Decussation : at the level of spinal cord

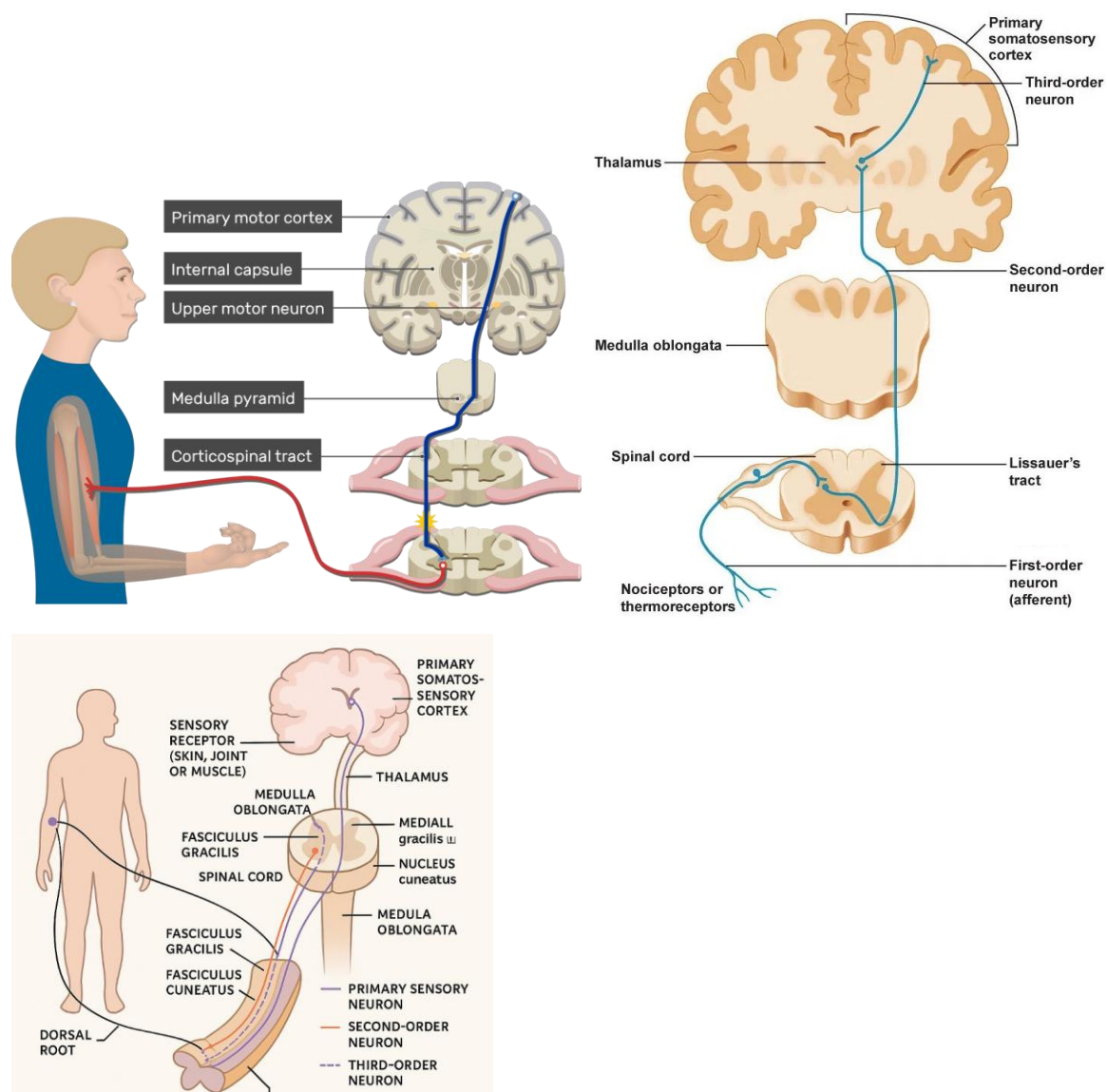
3.DORSAL COLUMN TRACT

Carries - fine touch, proprioception, vibration

First order neuron - DRG (dorsal root ganglion)

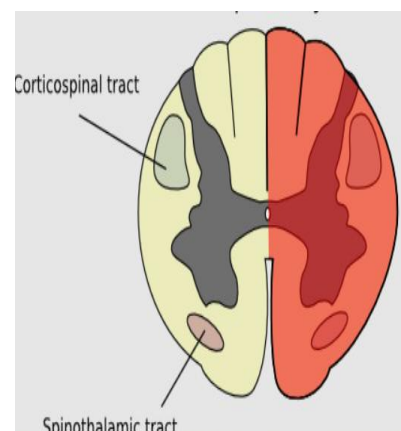
Second order neuron - gracilis (medially) and cuneatus (laterally) nucleus

Third order neuron - Thalamus

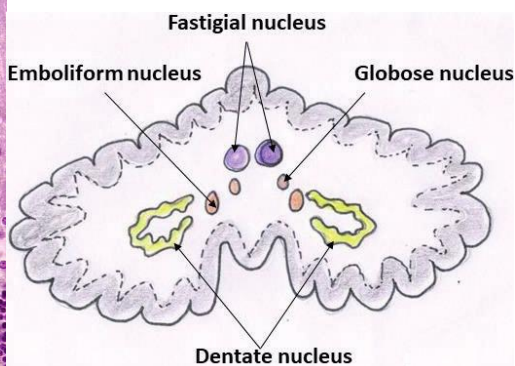
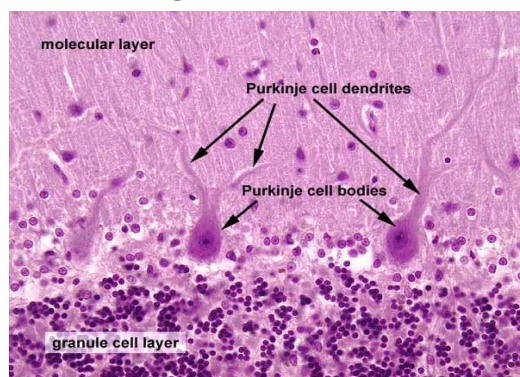


Hemisection of spinal cord

Signs of the Brown-Séquard syndrome	
Below the level of injury	Ipsilateral upper motor neuron paralysis Ipsilateral loss of conscious proprioceptive sensations Contralateral loss of pain and temperature and touch
At the level of injury	Ipsilateral lower motor neuron paralysis Ipsilateral anaesthesia over the skin of the segment
Above the level of injury	Ipsilateral hyperaesthesia



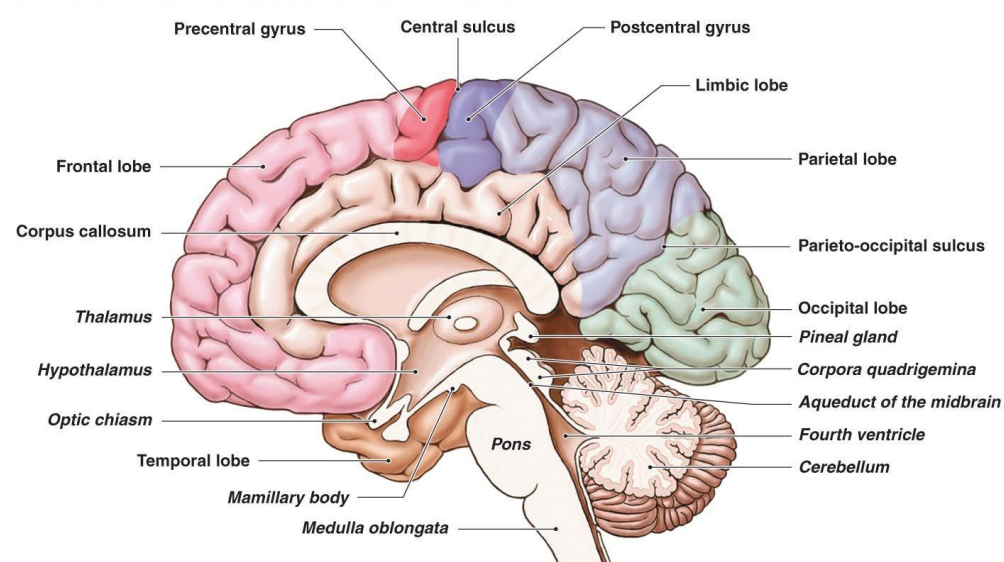
CEREBELLUM



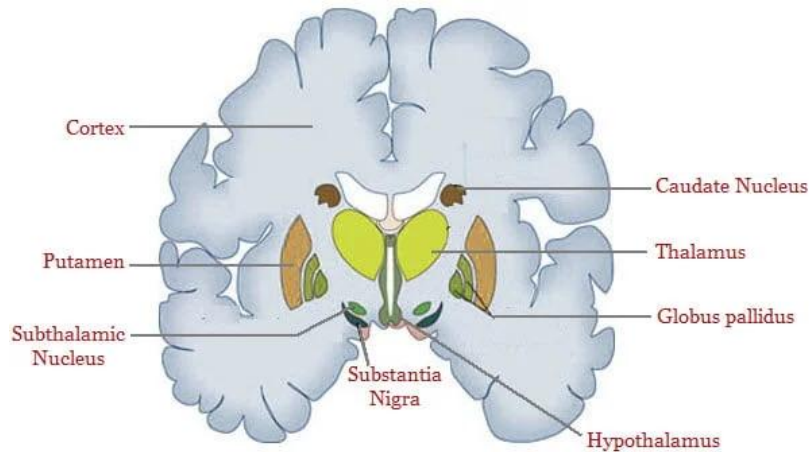
Lateral to medial → (trick : doctor eat good food)

Dentate nucleus → Emboliform → Globose → Fastigial

BRAIN



BASAL GANGLIA



LESION OF BASAL GANGLIA

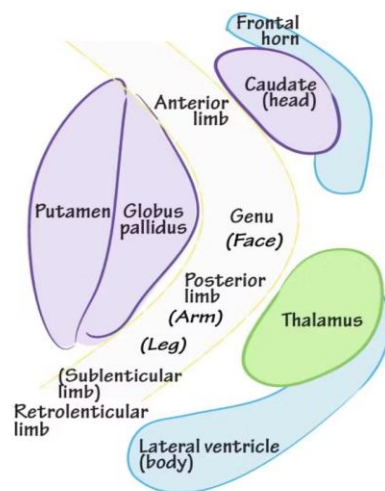
GLOBUS PALLIDUS → ATHETOSIS

PUTAMEN → CHOREA

SUBSTANTIA NIGRA → PARKINSON

SUBTHALAMUS → HEMIBALLISMUS

PARTS OF INTERNAL CAPSULE

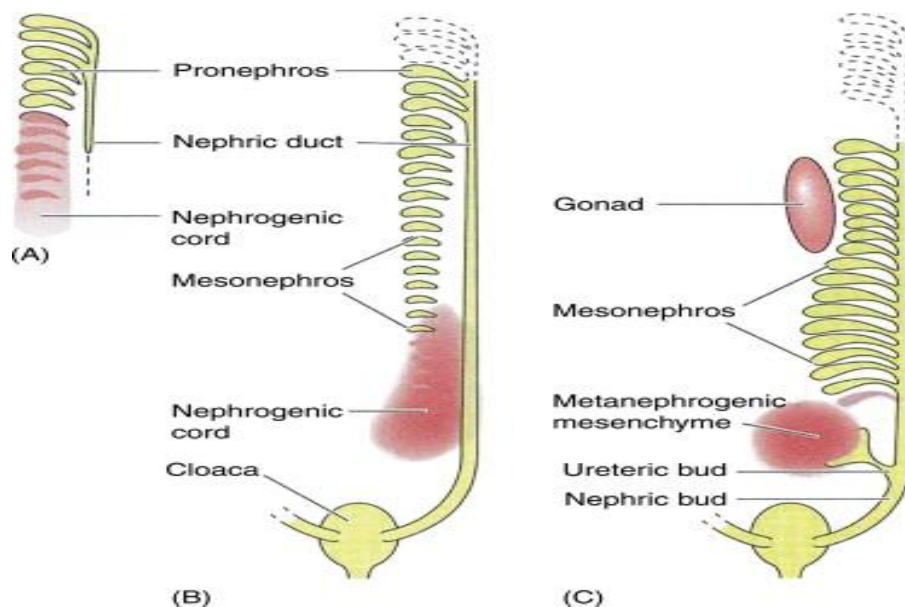


EMBRYOLOGY

Development of urinary bladder

Urinary bladder is derived from UGS (urogenital sinus) except trigone

Trigone derived from mesonephric duct



Q) Urogenital ridge develops from which layer?

Ans - Intermediate mesoderm

Q) PCT develops from ?

Ans - Metanephric Blastema

Q) ureteric bud develops from which duct ?

Ans - Mesonephric duct

Q) Trigone of posterior wall of urinary bladder formed by which structure ?

Ans - mesonephric duct

Male urethra derived from urogenital sinus except posterior wall of prostatic urethra
By mesonephric duct

Part of genital ducts	Derivatives/ remnants
Derivatives of mesonephric duct in male	• Ejaculatory duct • Seminal vesicle • Vas deferens • Efferent ductules of testis • Duct of epididymis
Vestigial remnants of mesonephric duct in male SLAP - trick	• Appendix of epididymis • Paradidymis • Superior aberrant ductules • Inferior aberrant ductules

Vestigeal remnants of paramesonephric duct in male	• Appendix of testis • Prostatic utricle
Fate of paramesonephric ducts in female	• Uterine tubes • Uterus • Upper part of vagina
Vestigeal remnants of mesonephric duct in female	• Epoophoron • Gartner's duct • Paroophoron
PEG - trick	

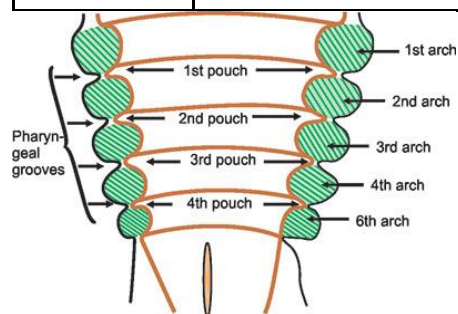
CLEFT	DERIVATIVES
1st	External auditory meatus, tympanic membrane
2nd-4th	Cervical sinus obliterate, persist as branchial cyst

POUCH	DERIVATIVES
1st	Middle ear cavity, eustachian tube, mastoid
2nd	Epithelial lining of crypts of palatine tonsil
3rd	Thymus, inferior parathyroid gland
4th	Superior parathyroid gland (ultimobranchial body- 5th remnant)

PHARYNGEAL ARCH ARTERIES

ARCHES	BONES	MUSCLES	NERVES	ARTERY
1st	Meckel cartilage, maxilla, mandible, malleus, incus, zygomatic bone	Muscles of mastication, mylohyoid, tensor veli palatini	V3	Maxillary artery
2nd	Reichert cartilage, stapes, styloid process, lesser horn of hyoid bone	Muscles of facial expression, stylohyoid, posterior belly of digastric	VII	Stapedial artery
3rd	Hyoid bone(body and greater horn)	stylopharyngeus	IX	Common carotid artery and internal carotid

4th	Laryngeal cartilages	Cricothyroid, levator palatini	X, superior laryngeal nerve	Left- aortic arch Right- proximal right subclavian artery
6th		All intrinsic muscles of larynx except cricothyroid	Vagus nerve, recurrent laryngeal nerve	Pulmonary artery, ductus arteriosus



Source of development of main arteries

Major arteries	Arteries Source of development
Arch of aorta	· Aortic sac (ventral part) · Left horn of aortic sac · Left fourth arch artery
Brachiocephalic artery	· Right horn of aortic sac
Right subclavian artery	· Proximal part from right fourth arch artery · Distal part from right seventh cervical intersegmental artery
Left subclavian artery	· Left seventh cervical intersegmental artery
Common carotid artery	· Third arch artery proximal to the external carotid artery bud
Internal carotid artery	· Third arch artery distal to the external carotid bud · Cranial part of dorsal aorta distal to the attachment of third arch artery
Descending aorta	· Proximal part from left dorsal aorta distal to attachment of fourth arch artery · Distal part from fused dorsal aortae forming single median artery

UMBILICAL LIGAMENT

Median- urachus

Medial- umbilical artery

Lateral- inferior epigastric artery

Left umbilical vein- ligamentum teres

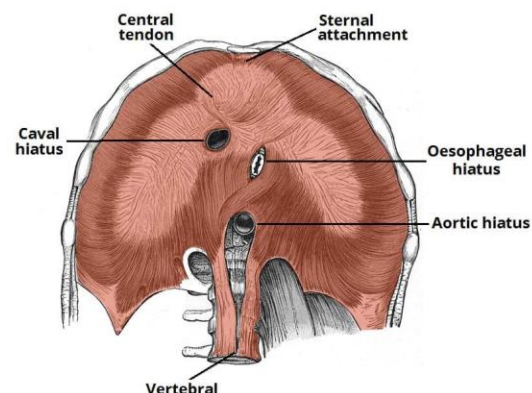
Structures piercing diaphragm

Mnemonic- I AT E(8) 10 eggs

IVC- T8-IVC, RIGHT PHRENIC NERVE

OESOPHAGUS- T10- esophagus, left gastric branch, vagus nerve

AORTIC- T12- azygous vein, aorta, thoracic



Development of diaphragm

Embryonic structures	Adult derivatives
Septum transversum	Central tendon of diaphragm
Pleuroperitoneal membranes	Small peripheral part of diaphragm
Dorsal mesentery of esophagus	Crura of diaphragm

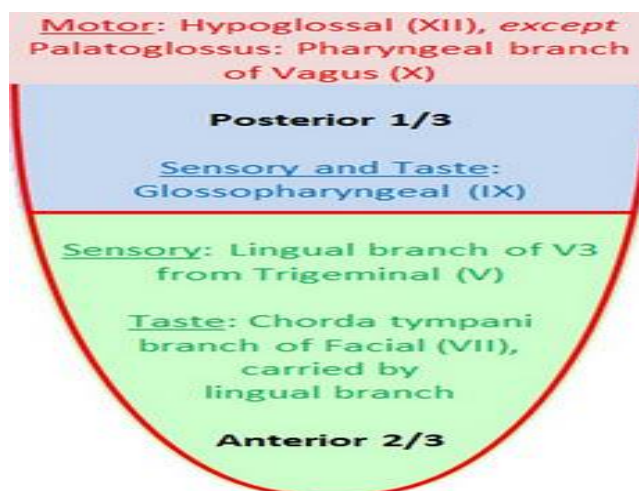
TONGUE

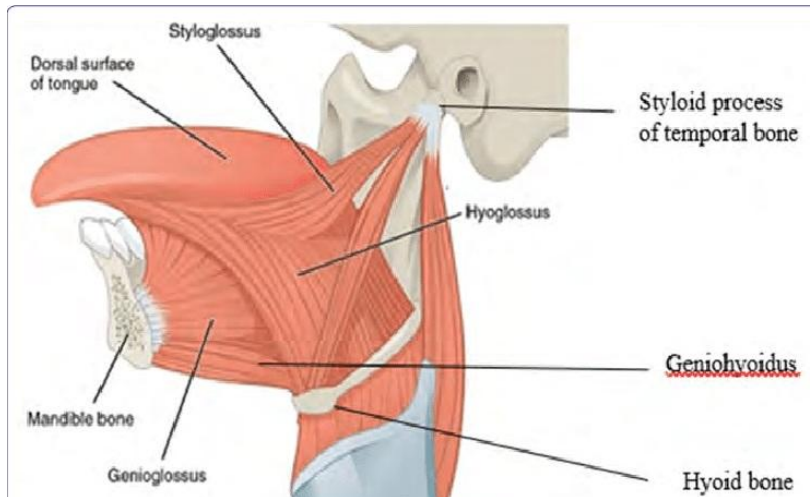
Q.All muscle supply by CN 12 except PALATOGLOSSUS- supply by IX/X

SAFETY muscle - GENIOGLOSSUS (prevent fall of tongue)

Depressor of tongue - HYOGLOSSUS

NOTE: when the patient is asked to protrude tongue, the tongue deviates to the paralysed side. That is the side of lesion of the ipsilateral hypoglossal nerve





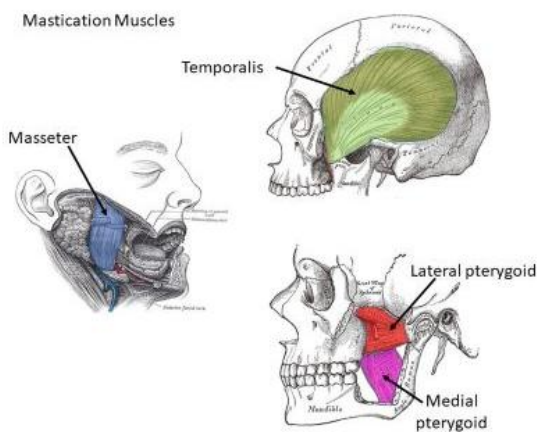
MUSCLE OF MASTICATION

All muscles are supplied by V3

Only retractor- TEMPORALIS

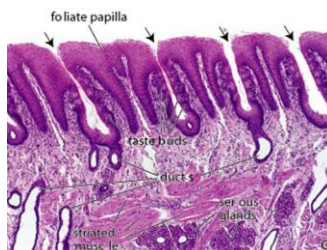
Depression of mandible/opening of jaw - LATERAL PTERYGOID (only depressor)

Most commonly involved in trismus - MEDIAL PTERYGOID

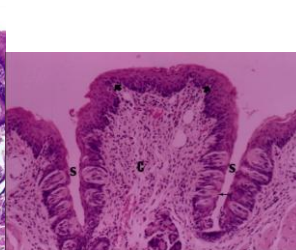


PAPILLAE OF TONGUE

Foliate



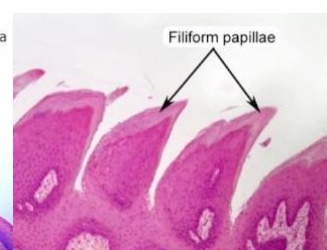
circumvallate



Fungiform



Filiform



PARASYMPATHETIC GANGLIA

GANGLIA	NERVE RELATED TO IT	STRUCTURE SUPPLIED BY IT
Ciliary ganglia	CN 3	Ciliaris + sphincter pupillae
Otic ganglia	CN 9	Parotid gland
Pterygopalatine ganglion	CN 7(greater petrosal nerve)	Lacrimal gland
Submandibular gland	CN 7(chorda tympani nerve)	Submandibular gland and sublingual gland

NERVE SUPPLY TO FACE

Sensory supply by - TRIGEMINAL NERVE except skin over MANDIBLE- supplied by GREAT AURICULAR NERVE

MOTOR SUPPLY OF FACE

All muscles are supplied by FACIAL NERVE except muscles of mastication- supplied by MANDIBULAR NERVE and LPS by oculomotor nerve.

UPPER LIMB**BRACHIAL PLEXUS****BRANCHES OF POSTERIOR CORD- ULTRA**

Upper subscapular nerve

Lower subscapular nerve

Thoracodorsal nerve

Radial nerve

Axillary nerve

BRANCHES OF MEDIAL CORD- M4U

Medial pectoral nerve

Medial root of median nerve

Medial cutaneous nerve of arm

Medial cutaneous nerve of forearm

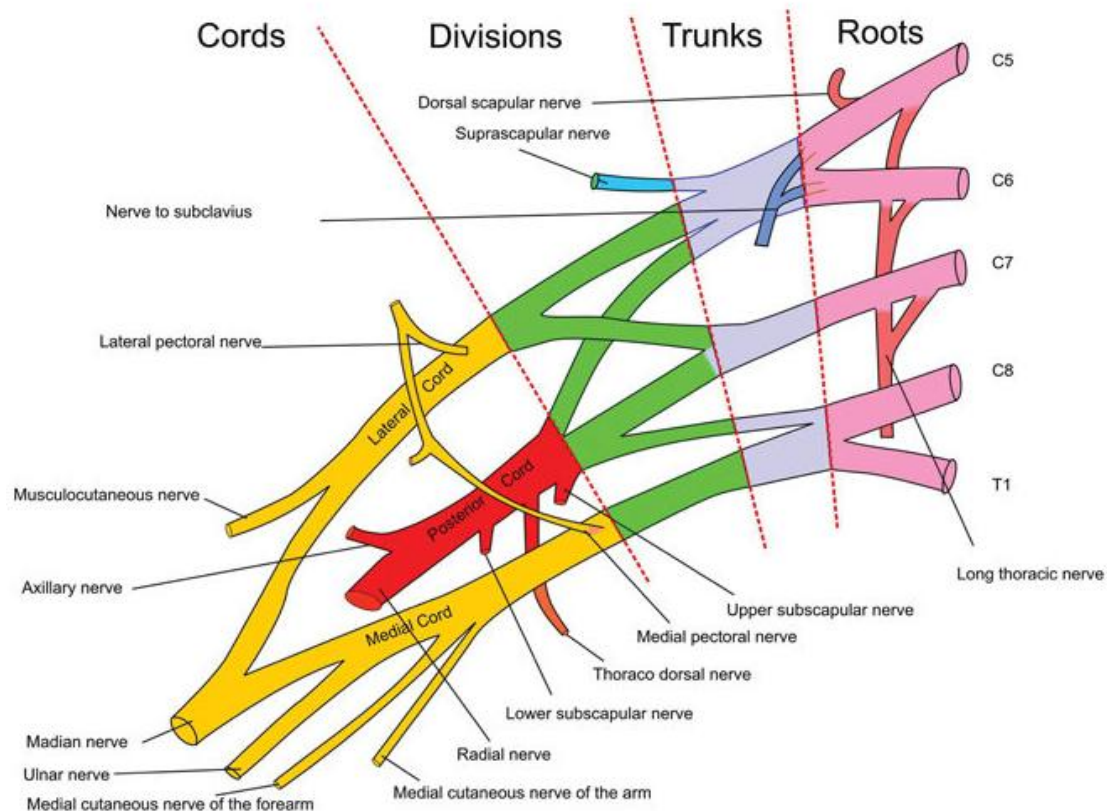
Ulnar nerve

BRANCHES OF LATERAL CORD - LML- Love Me Like u do

Lateral pectoral nerve

Musculocutaneous nerve

Lateral root of median nerve



SUBCLAVIAN ARTERY - mnemonic - VIT C D

1st part

V - Vertebral artery

I - Internal thoracic artery

T - Thyrocervical trunk -

S - Suprascapular artery

I - Inferior thyroid artery

T - Transverse cervical artery

A - Ascending cervical artery

2nd part

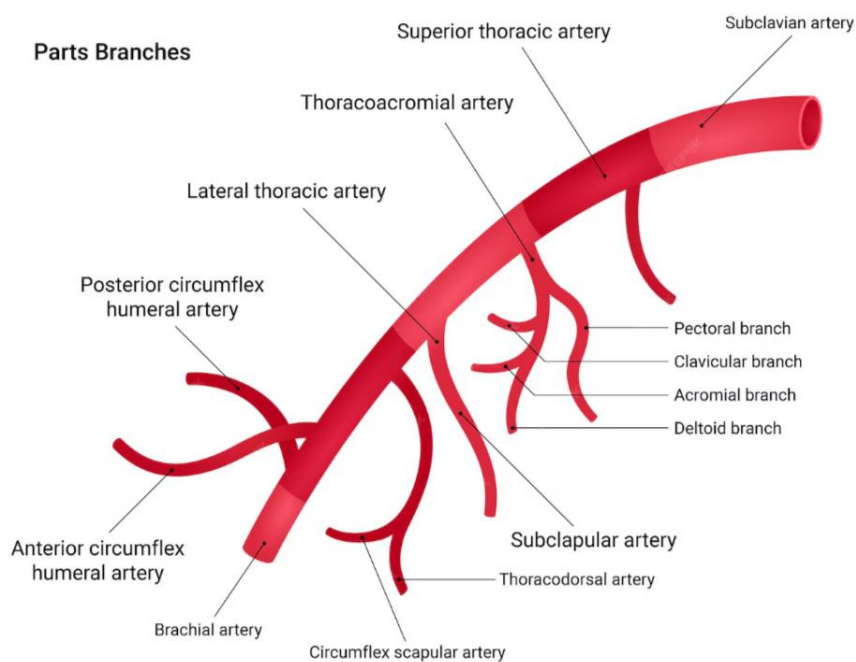
Superior intercostal artery

Deep cervical artery

3rd part

Dorsal scapular artery

AXILLARY ARTERY



CUBITAL FOSSA CONTENTS- MBBS

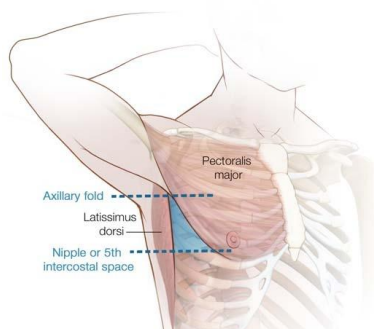
M- median nerve

B- brachial artery

B- tendons of biceps brachii

S- superficial radial nerve

Triangle of safety

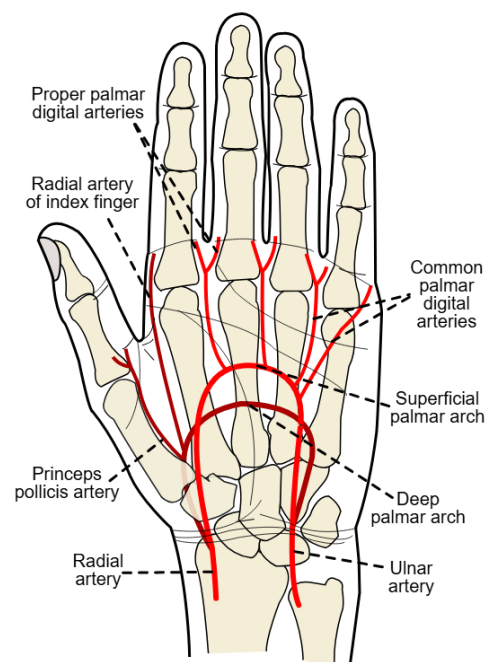


Lateral - latissimus dorsi

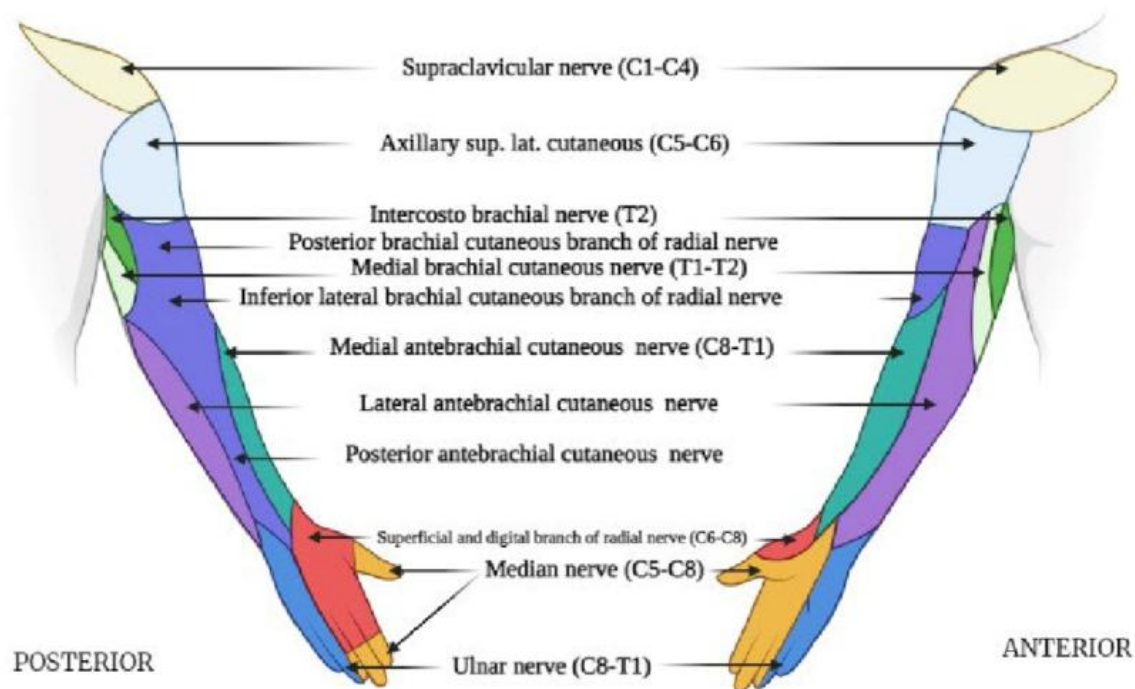
Medial - pectoralis major

Inferior - horizontal line from nipple

PALMAR ARCH ARTERIES



DERMATOME



RIBS

True ribs - 1 to 7

False ribs - 8,9,10

Floating ribs - 11,12

Neurovascular bundle - at lower border of ribs

V- intercostal vein

A- intercostal artery

N- intercostal nerves

ONE LINERS

Cranial nerve which carry parasympathetic fibers- 3,7,9,10

Thickest/largest CN - TRIGEMINAL NERVE

Most thinnest CN - TROCHLEAR NERVE CN

which undergoes INTERNAL DECUSATION - TROCHLEAR NERVE

Crossing CN with longest intracranial course - TROCHLEAR NERVE

CN with longest extra cranial course - VAGUS NERVE

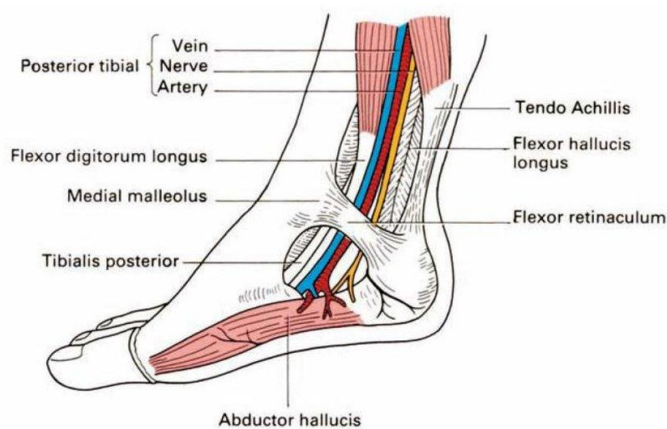
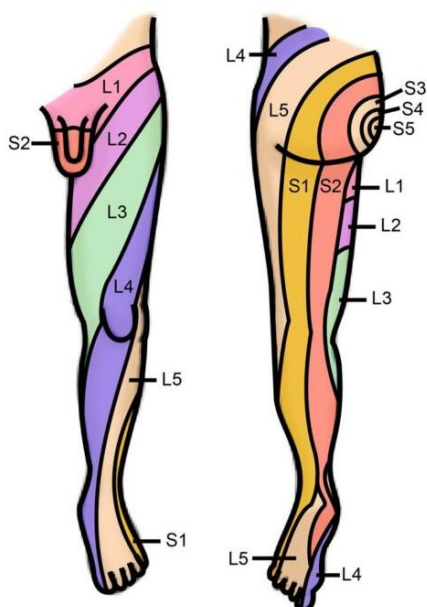
CN with longest intraosseous course - FACIAL NERVE

CN involved in increased ICP- ABDUCENS NERVE

LOWER LIMB

Highest point of iliac crest- L4

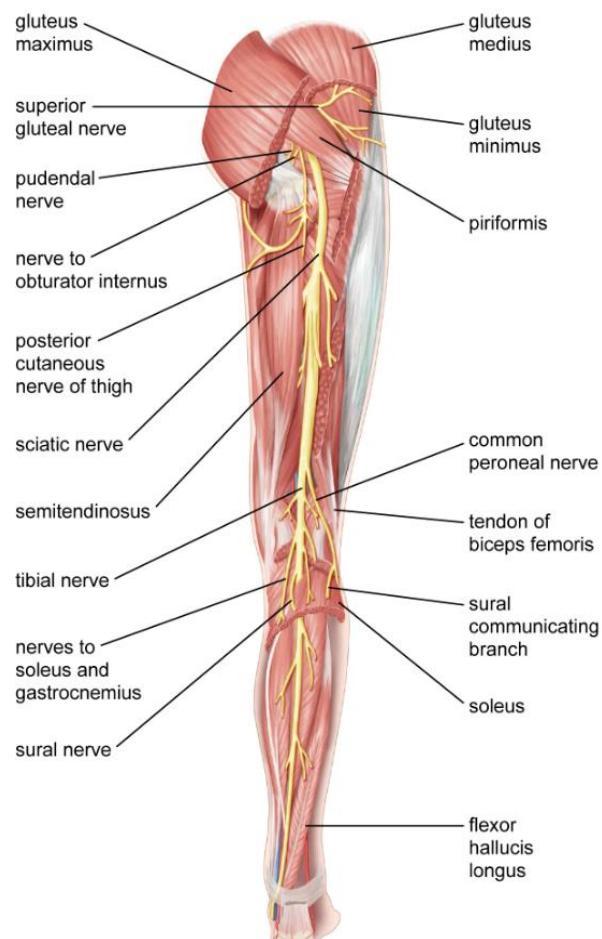
LOWER LIMB DERMATOME

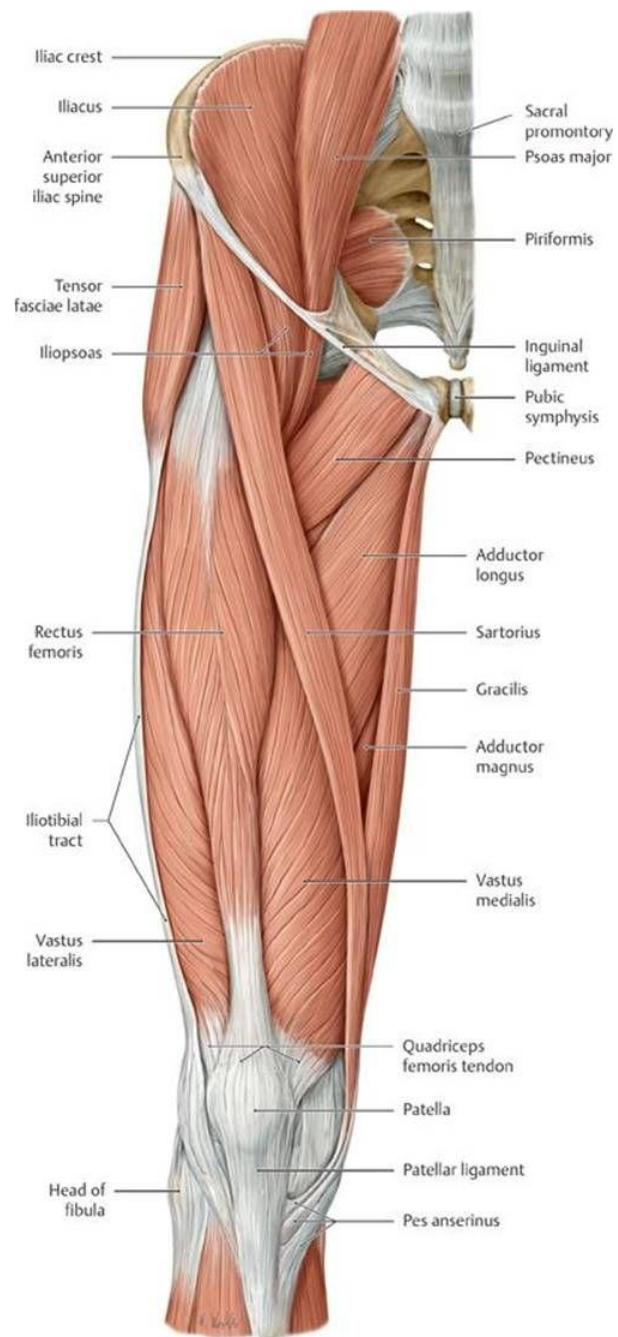


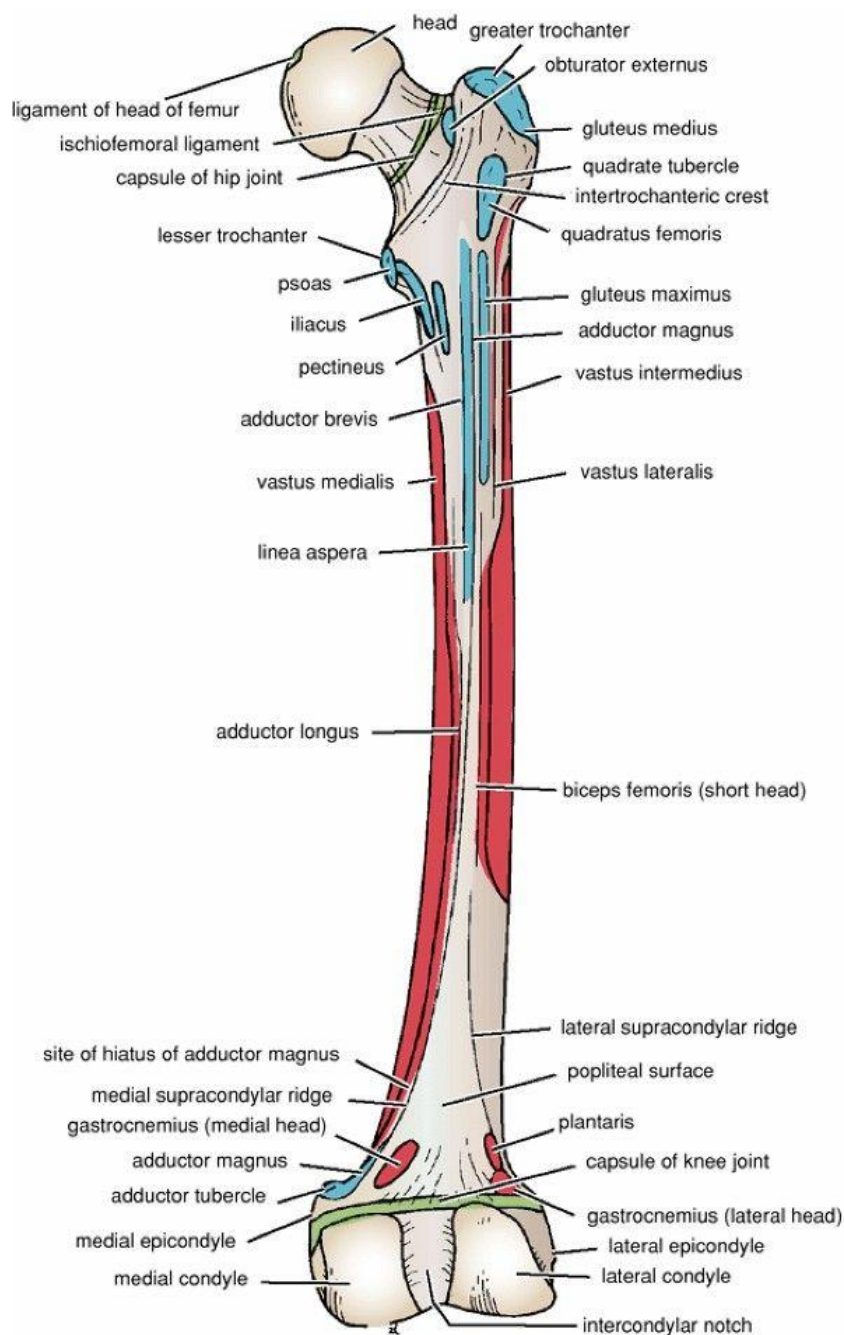
MOST IMP TABLE

Muscles	Segment	Nerve
ILIOPSOAS	L1-L4	FEMORAL
QUADRICEPS FEMORIS	L2-L4	FEMORAL
GLUTEUS MEDIUS AND MINIMUS	L2-L4	OBTURATOR
GLUTEUS MAXIMUS	L5-S2	SUPERIOR GLUTEAL

BICEPS FEMORIS, SEMITENDINOSUS, SEMIMEMBRANOSUS	L5-S2	SCIATIC
EXTENSOR HALLUCIS LONGUS	L5	DEEP FIBULAR
EXTENSOR DIGITORUM LONGUS, BREVIS	L5-S1	DEEP FIBULAR







STOMACH

Blood supply - Mnemonic - "Left Right kids play Soccer"

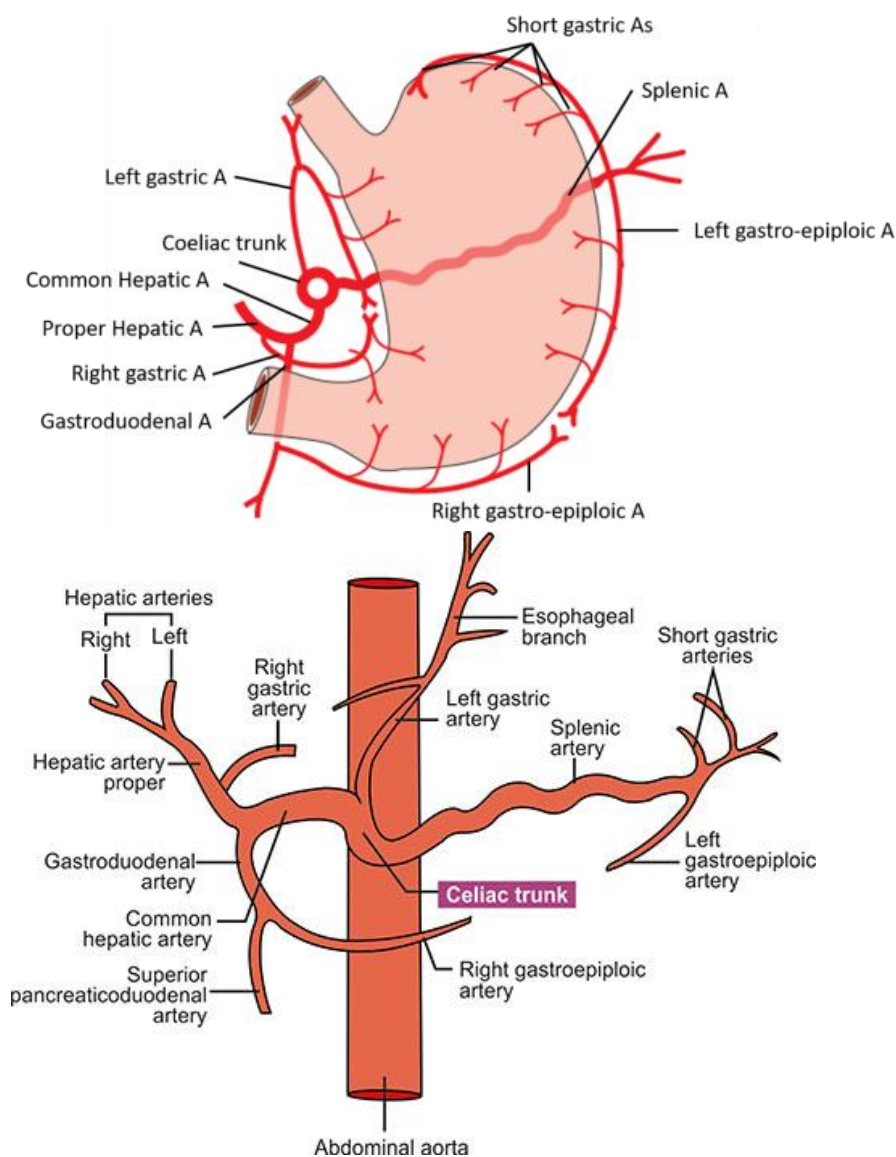
Left gastric artery - supplies the lesser curvature of the stomach (arises from the celiac trunk).

MC artery to bleed in GI bleed Right gastric artery - supplies the lesser curvature of the stomach (arises from the hepatic artery proper).

Left gastroepiploic artery - supplies the greater curvature of the stomach (arises from the splenic artery).

Right gastroepiploic artery- supplies the greater curvature of the stomach (arises from the gastroduodenal artery).

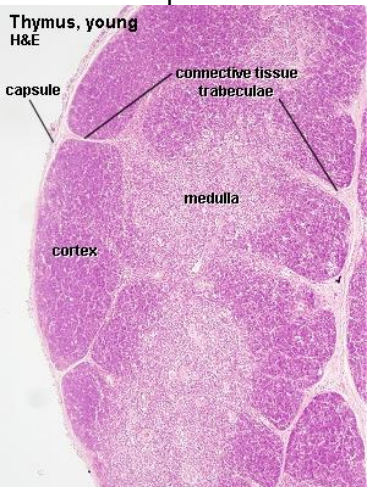
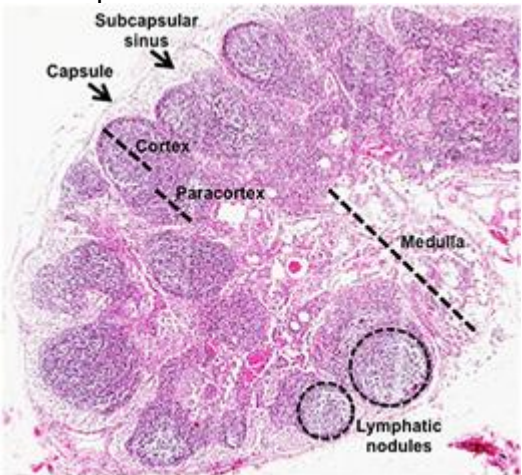
Short gastric arteries- supply the fundus of the stomach (arise from the splenic artery).

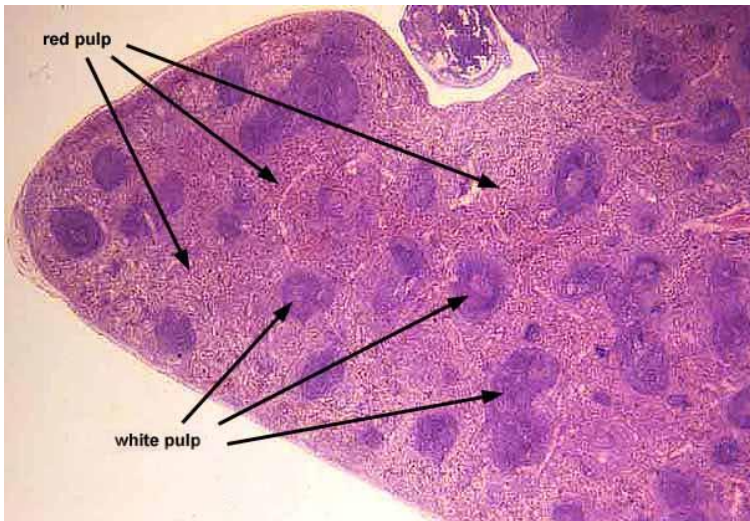
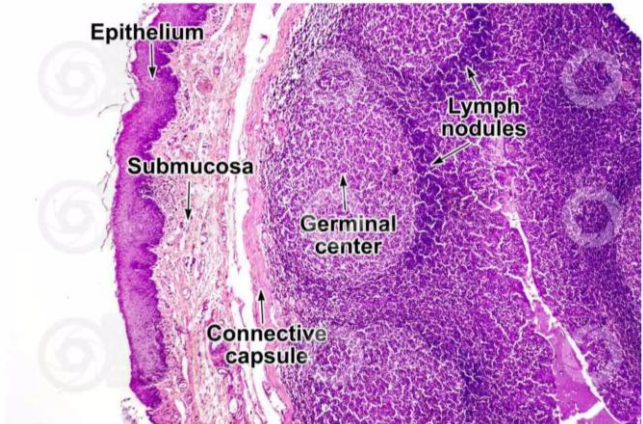


GREATER SCIATIC FORAMEN	LESSER SCIATIC FORAMEN
PINS- P3 I N S2	IPO
Pudendal nerve Posterior femoral cutaneous nerve Piriformis muscle Inferior gluteal nerve and vessels Nerve to obturator internus Sciatic nerve Superior gluteal nerve and vessels	Internal pudendal vessels Pudendal nerve Obturator internus tendon

Note- Only remember lesser sciatic foramen

Lymphoid tissue

Organ	Identifying features
Thymus	Many lobules of lymphoid tissue Darkly stained cortex and lightly stained medulla in each lobule Hassall's corpuscles in the medulla  Thymus, young H&E capsule connective tissue trabeculae medulla cortex
Lymphnode	Lymphatic nodules in the cortex Subcapsular sinus Medullary cords and sinuses in the medulla Thin capsule  Subcapsular sinus Capsule Cortex Paracortex Medulla Lymphatic nodules
Spleen	White pulp (lymphatic nodule) transversed by central arteriole Red pulp containing splenic cords and sinusoids trabeculae Thick capsule and trabeculae

	 red pulp white pulp
Palatine tonsil	Crypts lined by stratified squamous epithelium Sub epithelial lymphatic nodule Mucous glands  Epithelium Submucosa Connective capsule Lymph nodules Germinal center

Remnants of fetal structures after birth

Embryonic structure	Remnants in adult
Vitelointestinal duct	Meckel's diverticulum
Allantois	Median umbilical ligament
Umbilical artery	Superior vesicle artery Medial umbilical ligament
Left umbilical vein	Ligamentum teres hepatitis